

Chapter 5 - Matrices and Determinants

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#Matrix

Matrix is a rectangular array of numbers arranged in rows and columns which are enclosed inside brackets.

E.g., $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 3 \end{pmatrix}$, etc.

Order of Matrix

An order of a matrix is the size of matrix represented as $(r \times c)$ where r is no. of rows and c is no. of columns.

#Special Types of Matrices

1. Square Matrix

A square matrix is a special matrix having the same number of rows and columns.

E.g., $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 3 \end{pmatrix}$, etc.

2. Unit/Identity Matrix

Unit/Identity matrix is the square matrix whose diagonal elements are 1 and all other elements are zero.

E.g., $I_1 = (1)$, $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, etc.

3. Zero/Null Matrix

Zero/Null matrix is the matrix whose all elements are zero.

E.g., $0_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, etc.

4. **Triangular Matrix**

Triangular matrix is the square matrix whose all elements either below or above the diagonal are zero.

There are two types of triangular matrix:

a. **Upper Triangular Matrix**

Upper triangular matrix is the square matrix whose all elements below the diagonal are zero.

$$\text{E.g., } \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & -3 \end{pmatrix}, \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix}, \text{ etc.}$$

b. **Lower Triangular Matrix**

Lower triangular matrix is the square matrix whose all elements above the diagonal are zero.

$$\text{E.g., } \begin{pmatrix} 5 & 0 \\ 3 & \sqrt{2} \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ -3 & -1 & -2 \end{pmatrix}, \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \text{ etc.}$$

5. **Symmetric Matrix**

Symmetric matrix is the square matrix such that $a_{ij} = a_{ji}$ for all i, j or $A = A^T$.

$$\text{E.g., } \begin{pmatrix} 1 & 3 & 7 \\ 3 & 0 & -6 \\ 7 & -6 & 8 \end{pmatrix}$$

6. **Skew-symmetric Matrix**

Skew-symmetric matrix is the square matrix such that $a_{ij} = -a_{ji}$ for all i, j or $A = -A^T$.

$$\text{E.g., } \begin{pmatrix} 0 & 5 & -7 \\ -5 & 0 & 3 \\ 7 & -3 & 0 \end{pmatrix}$$

#Transpose of a Matrix

Transpose of a matrix is the new matrix obtained by interchanging the rows and columns of the original matrix.

Transpose of a matrix A is denoted by A^T or A' .

Properties of Transposed Matrices

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(kA)^T = kA^T$
4. $(AB)^T = B^T A^T$
5. $A + A^T \rightarrow$ symmetric matrix
6. $A - A^T \rightarrow$ skew-symmetric matrix
7. $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

#Determinant

Determinant is a special scalar value of the geometric quantity interpreted/scaled by a square matrix in the dimension of its order. i.e., In 1D, it gives the length,

2D, it gives the area,

3D, it gives the volume. Determinant of matrix A is denoted by $\det(A)$ or $|A|$.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Minor and Cofactor

Minor of element a_{ij} is the determinant of a matrix formed by removing the i^{th} row and j^{th} column.

Minor of element a_{ij} is denoted by M_{ij} . E.g., Minor of element a_{12} is $\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

Cofactor of element a_{ij} is the product of $(-1)^{i+j}$ and minor of element a_{ij} .

Cofactor of element a_{ij} of matrix A is denoted by A_{ij} . E.g., Cofactor of element a_{12} is $(-1)^{(1+2)} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

Sarrus Rule

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Using Sarrus Rule,

$$\begin{array}{ccccccc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} & & \\ & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} & & \\ & \swarrow & \searrow & \swarrow & \searrow & \swarrow & \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} & & \end{array} \begin{array}{l} - \\ + \end{array}$$

$$= (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12})$$

Properties of Determinants

There are certain properties of determinants of a matrix by which their evaluation becomes much easier:

1. The determinant of a matrix is not affected by interchanging rows and columns.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

2. Interchanging any two adjacent rows/columns negates the determinant.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = - \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

3. If any two rows/columns of a matrix are identical, its determinant is zero.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

4. If a row/column of a matrix is multiplied by a scalar, we can bring it outside the determinant.

$$\begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

5. If a row/column is the sum of two terms, we can distribute the determinant in sum.

$$\begin{vmatrix} a_1 + \alpha & b_1 & c_1 \\ a_2 + \beta & b_2 & c_2 \\ a_3 + \gamma & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} \alpha & b_1 & c_1 \\ \beta & b_2 & c_2 \\ \gamma & b_3 & c_3 \end{vmatrix}$$

6. The determinant is not affected if a row/column is added with a scalar times other row/column.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 + ka_2 & b_1 + kb_2 & c_1 + kc_2 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Special Case: $k = \pm 1$

The determinant is not affected if a row/column is added/subtracted with another row/column.

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 \pm a_2 & b_1 \pm b_2 & c_1 \pm c_2 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

#Singular and Non-Singular Matrix

Singular Matrix: Singular matrix is the square matrix whose determinant is zero.

Non-Singular Matrix: Non-singular matrix is the square matrix whose determinant is not zero.

#Adjoint/Adjugate of a Matrix

An (adjoint or adjugate) of a matrix is the matrix formed by transposition of its cofactor matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \Rightarrow \text{Adj}(A) = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}^T = \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$$

#Inverse of a Matrix

Inverse of a matrix is the matrix that gives identity matrix when multiplied with the original matrix.

A non-singular matrix only has an inverse matrix. The inverse of a matrix A is denoted by A^{-1} .

Property

$$AA^{-1} = A^{-1}A = I$$

Relation with Adjoint Matrix

$$A^{-1} = \frac{\text{Adj}(A)}{|A|}$$

where I is the identity matrix of the order of A .